



INDIAN INSTITUTE
OF TECHNOLOGY
PALAKKAD

DEPARTMENT OF MATHEMATICS

IIT PALAKKAD

ABOUT THE DEPARTMENT

Department of Mathematics at IIT Palakkad is strongly committed to excellence in research and teaching. We envisage a period of fast growth leading to a strong and diverse Mathematics community at IIT Palakkad. The expertise of the faculties spread across pure and applied mathematics. The department offers a doctoral research program from 2017 for students who would like to pursue their career in mathematics. The department is very keen on having researchers in every major field of mathematics and are proactive to ensure that this is achieved in the near future. In addition, the PhD students are actively involved as teaching assistants for both B. Tech as well as MSc courses. This will prepare them well for a future career in academia.

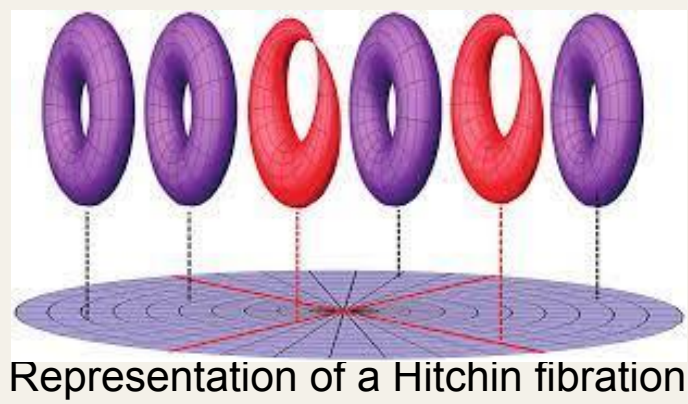
PROGRAMS

- **M.Sc. Mathematics** : The MSc Mathematics program provides students with a strong mathematical foundation and also offers an opportunity to specialize in diverse areas such as pure mathematics, applied mathematics, data sciences and theoretical computer science. The curriculum has 12 core courses and 8 electives which gives the students plenty of freedom to focus on the areas of their interest. Students will also have the option to pursue a project in place of some of the electives, helping them to gain hands-on training in a specific area. This helps students pursue careers in both academia and industry.
- **MTech Programs** : The department is also part of two interdisciplinary programs: MTech in Computing and Mathematics and MTech in Data Science. Students who join these programs take several core master's level courses offered by the mathematics department.
- **PhD**: Students who pursue a PhD at IIT Palakkad get an opportunity to work in a wide range of fields in both pure and applied mathematics. The young and vibrant faculty have ongoing collaborations with renowned researchers both in India as well as abroad. This gives PhD students in the department ample scope to work with, and learn from, some of the foremost experts in the relevant fields. The department is very keen on having researchers in every major field of mathematics and are proactive to ensure that this is achieved in the near future.
- **BTech (All branches)** : The department is actively involved in the BTech programs of the institute. Several core mathematics courses such as Calculus, Linear Algebra and elective courses such as differential equations Probability and Statistics, Algebra, Complex Variables, Numerical Analysis are also offered by the department.

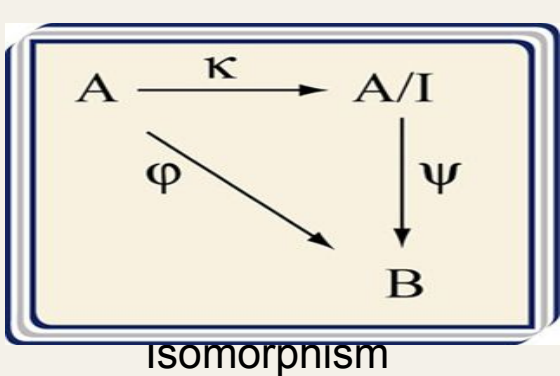
RESEARCH

The research expertise of the present faculty members are the following

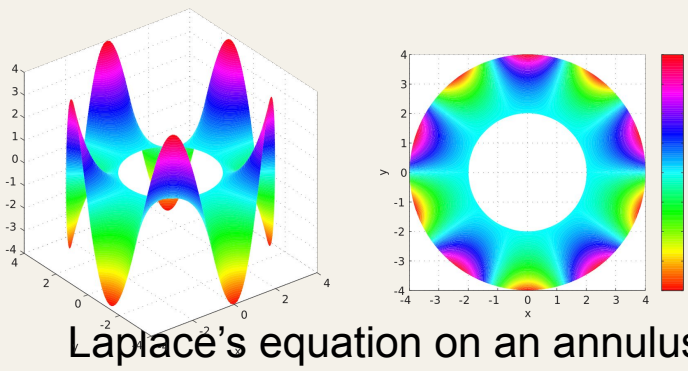
•Algebraic Geometry



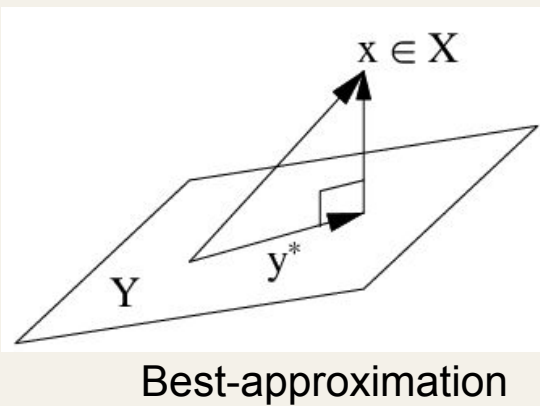
•Commutative Algebra



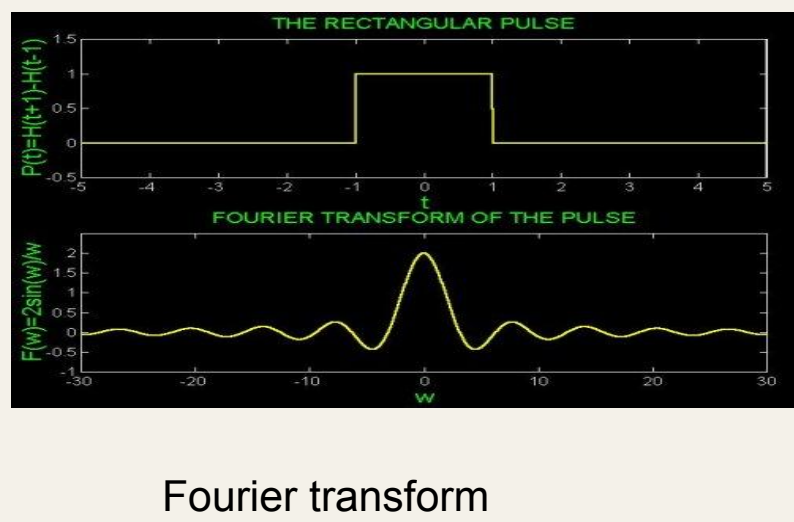
•Differential Equations



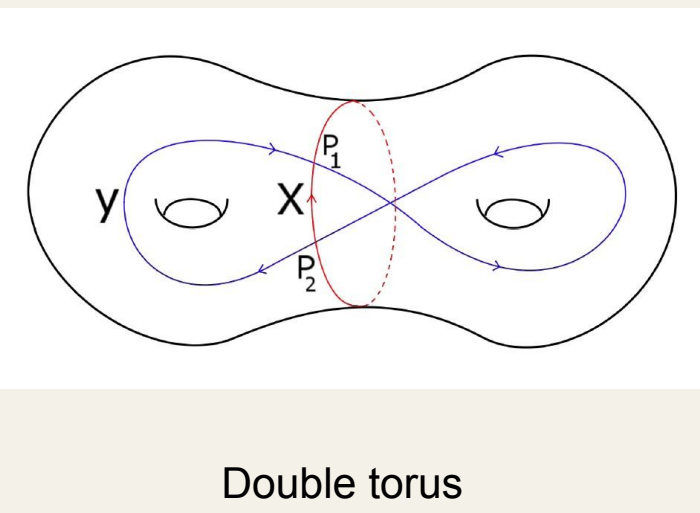
•Functional Analysis



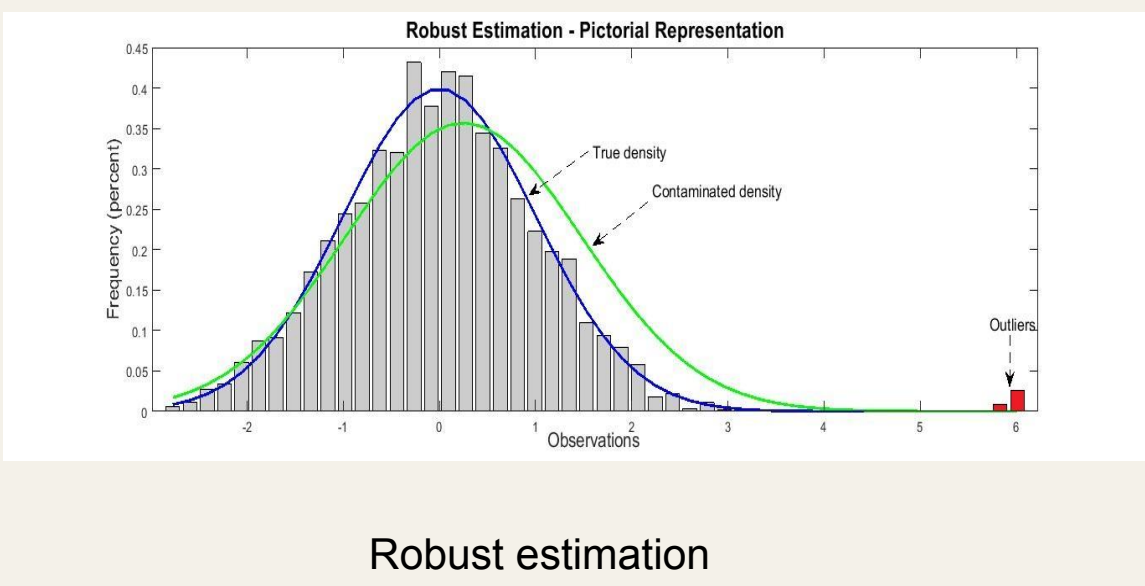
•Harmonic Analysis



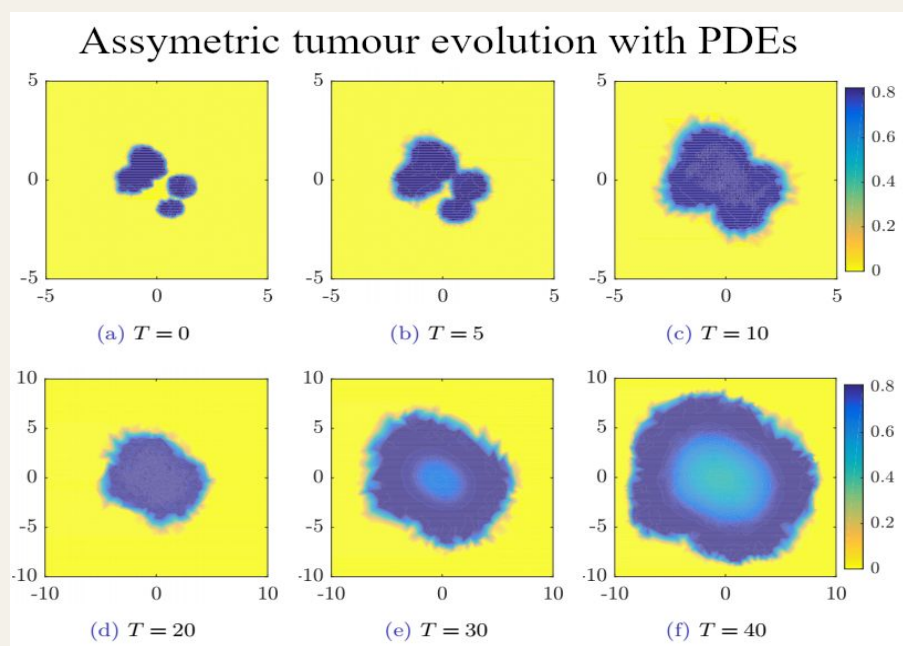
•Low-dimensional Topology



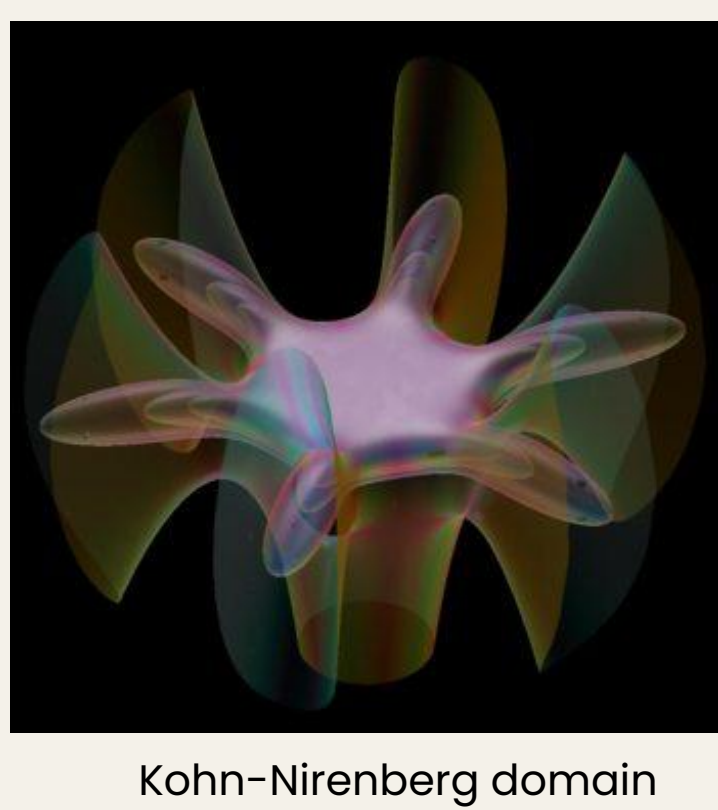
•Mathematical Statistics, Probability and Information Theory



• Numerical Analysis and Mathematical Biology



•Several Complex Variables



DEPARTMENTAL ACHIEVEMENTS

- Dr Jayanarayanan won **Kerala State Young Scientist Award 2021**.
- Dr Lakshmi Sankar: **SERB Core Research Grant (CRG)**, Project Title : Semilinear Elliptic Boundary Value Problems on Unbounded Domains in $\mathbb{R}^2$ (for three years starting from 2023).
- Dr Arpan Kabiraj: **SERB Start-up Research Grant (SRG)**. Project Title: Study of Skein Algebras Through Quantization of Poisson Algebras (for two years starting from 2023).
- Dr Arpan Kabiraj: **SERB Mathematical Research Impact Centric Support (MATRICS)**. Project Title: The Study of Turaev Cobracket from a Hyperbolic Geometric Perspective (for three years starting from 2023).
- Dr Ashok Kumar: **SERB Mathematical Research Impact Centric Support (MATRICS)**. Project Title: Information Geometric Approach to Estimation Error Bounds (for three years starting from 2023).
- Dr Parangama Sarkar: **SERB-POWER Research Grant**. Project Title: Multiplicities and Graded Algebras Associated to Filtrations of Ideals (for three years starting from 2022).
- Dr Jayanarayanan: **SERB Mathematical Research Impact Centric Support (MATRICS)**. Project Title: Study of Approximation Theoretic Properties of the Space of Compact Operators on Banach Spaces (completed in 2021).

ACTIVITIES

- The department regularly organises a seminar series called 'Yuktibhasa Seminar Series'. These are expository seminars aimed to be accessible by mathematics enthusiasts familiar with mathematics upto undergraduate level. Recordings of these seminars which took place in the online mode can be viewed from the respective webpage of the department.
- The department also organises 'Department Seminar Series' regularly where faculty members and students of the department deliver talks.
- The department actively engages in outreach activities such as 'MathCircle'.
- The department organises Departmental Symposium once every year.

STRENGTH

- Number of Faculty Members : 12
- Number of Research Scholars : 19
- Number of M.Sc. Mathematics 1st Year (2023 batch) : 22
- Number of M.Sc. Mathematics 2nd Year (2022 batch) : 22

CONTACT

Department of Mathematics
(Email : office_ma@iitpkd.ac.in)
Indian Institute of Technology Palakkad,
Ahalia Integrated Campus,
Kozhippara P. O | Palakkad,
Kerala | Pin: 678557